

Concept Architecture · Version 1.0

Universal Security & Privacy Framework

Simple Bhasha Mein Poori Samajh

Kisi bhi communication app ke liye ek privacy design —
jahan user khud decide kare uska data kab dikhe, kisko
dikhe, aur kitna dikhe.

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1. Framework Ka Maqsad

Ye framework kisi bhi communication app mein integrate kiya ja sakta hai — chahe wo app **Offline** ho, **Online** ho, ya **Hybrid** ho.

Sabse zaroori baat ye samajhna hai ki iska maqsad **sirf encryption karna nahi hai**. Encryption to ek zariya hai — asli maqsad hai user ko poora control dena ki uska data kab, kisko, aur kitna dikhe.

Framework in cheezo par khada hai:

- **User Controlled Privacy** — data ka har faisla user khud leta hai
- **Multi-Level Security** — zarurat ke hisaab se security ka level chuna ja sakta hai
- **Mutual Access Control** — sensitive cheezein sirf dono logon ki raza (permission) se khulti hain
- **Hidden Data Protection** — kuch data bilkul chhupa reh sakta hai, kisi ko uske hone ka pata bhi nahi chalta
- **Plausible Deniability** — agar koi zabardasti data dikhwaye, to user ke paas ek aisa layer hota hai jisse asli sensitive data ka wajood hi saamne nahi aata
- **Decoy Based Privacy** — fake (nakli) data dikha kar real data ko bachaya jata hai

2. Do Security Modes

Framework do modes deta hai — user apni zarurat ke hisaab se koi ek chun sakta hai.

Normal Mode	Advanced Mode
Rozmarra (daily) use karne wale users ke liye	High privacy chahne wale users ke liye
Sirf 2 protected folders: Locked aur Hidden	Three Level Security, Decoy System, Recursive Privacy, sab available
Security sirf user ke password par based hai	Password + dusre user ki permission + device verification tak ja sakti hai
Simple aur fast	Zyada layers, zyada control, thoda zyada setup

3. Normal Mode: Locked Folder & Hidden Folder

Locked Folder

User apne password se protect karta hai. Sirf owner (jisne folder banaya) hi ise access kar sakta hai — bilkul kisi bhi normal locked app jaisa.

Hidden Folder

Ye folder normal locked folder se alag hai — isliye isko khaas tareeke se banaya gaya hai:

- Password bhi lagta hai, lekin folder khud kahin dikhta nahi
- Isko kholne ke liye Search Bar mein password daal kar Search Button **3 baar** tap karna hota hai
- Iska koi icon ya button app mein kahin nahi hota

ISSE FAYDA KYA HAI

Agar aapke kisi normal chat ka naam bhi Hidden Folder ke password jaisa hai, tab bhi wo normal search se Hidden Folder ko accidentally reveal nahi karega — kyunki Normal Search aur Hidden Search ka process poori tarah alag rakha gaya hai.

Normal Mode mein Three Level Security aur Decoy System nahi hote — sirf seedha password-based protection hoti hai, taaki use karna simple rahe.

4. Advanced Mode: Ek Nazar Mein

Advanced Mode un users ke liye hai jinhe apni baatcheet ki privacy par sabse zyada control chahiye. Isme ye sab milta hai:

- **Three Level Security** — normal se lekar bohot high-trust security tak
- **Hidden Workspace** — chhupi hui, na-dikhne wali jagah
- **Decoy System** — ek hi jagah, alag password se alag data
- **Recursive Privacy** — layer ke andar layer
- **Mutual Authentication** — dono logon ki sehmati se hi sensitive access khulta hai

Aage ke sections mein har feature ko example ke saath samjhaya gaya hai.

5. Three Level Security

Ye teeno level Visible Chat aur Hidden Chat — dono par equally lagu hote hain.

Level	Authentication	Purani Chat Kab Dikhti Hai	Kis Ke Liye Best
Level 1	Sirf apna password	Turant, hamesha	Roz ki normal baatcheet, jahan purani chat baar-baar dekhni ho
Level 2	Apna password	Sirf jab doosra user permission de	Live secure conversation — jahan sirf abhi ki baat matter karti ho
Level 3	Apna password + doosre ki approval + Local Device Verification Code	Sirf jab doosra allow kare, aur dono devices paas-paas hon	Bohot sensitive data, jahan ek doosre par poora bharosa bhi na ho

Level 1 — Roz Ki Baatcheet

User apna password daal kar chat khol sakta hai. Purani chat kabhi bhi dekhi ja sakti hai, aur pending (bina padhe) messages normal tarike se dikhte rehte hain. Ye sabse simple level hai.

Level 2 — Live Secure Conversation

User apna password daal kar folder mein to ghus sakta hai, lekin purani chat turant nahi dikhegi — jab tak doosra user ijazat na de. Agar doosra user offline hai aur usse message bheje gaye hain, to wo pending messages bhi encrypted state mein "Old Chat" section mein store ho jaate hain, jab tak wo online aakar dekhne ki permission na de.

KAB USE KAREIN

Agar आपको बार-बार पुरानी chat देखनी पड़ती है, तो Level 1 ज्यादा suitable रहेगा — Level 2 खास तौर पर "sirf abhi ki baat" wali secure conversations के liye बना है।

Doosre User Ki Approval — Do Tareeke

Old chat dekhne के liye doosre user से approval लेने के दो तरीके हो सकते हैं। User इन दोनों में से कोई एक चुन सकता है, या दोनों तरीके एक साथ भी use हो सकते हैं।

TAREEKA 1 — LIVE CONFIRM REQUEST

User old chat dekhne के liye upar बने **locked box** पर click करता है। Isse doosre user के पास एक confirm request जाती है। Ye request tabhi travel करेगी जब dono users उस waqt **live (online)** हों — अगर dono में से कोई भी एक offline है, तो request travel नहीं करेगी, वो fail हो जायेगी। जैसे ही dono live होते हैं, request भेज दी जाती है — और doosra user जब **Confirm** पर click करता है, tabhi requesting user की locked old chat खुलती है।

TAREEKA 2 — PASSWORD SE

Agar user इस तरीके से old chat dekhna चाहे, तो doosre user का password भी दालना पड़ता है। Password batana या ना batana पूरी तरह doosre user की अपनी मर्जी पर है। जब तक dono live हैं, doosra user अपना password कभी भी बदल सकता है — ताकि अगर किसी को एक बार password बताया गया हो, तो वो उसी पुराने password से दोबारा access ना ले सके।

IS TAREEKE MEIN RISK

Password वाले तरीके में एक risk हमेशा रहता है — password पता चले ही, doosra user खुद data expose कर सकता है, या हो सकता है उसका device ही कोई और चला रहा हो (सही password का मतलब ये नहीं कि उस तरफ वही asli insaan बैठा है)। इस risk को कम करने के liye Level 3 में एक extra shart जुड़ी है — device का physically पास होना और Local Device Verification Code (नीचे बताया गया है)। Isse sirf sahi password काफी नहीं रहता — sahi device भी वहाँ physically मौजूद होना चाहिए।

ये दोनों तरीके — Live Confirm Request और Password — Level 2 और Level 3, दोनों में काम करते हैं। Level 3 में बस इतना extra है कि इनके साथ-साथ device का near होना और Offline Verification Code भी जरूरी हो जाता है।

Level 3 — Jab Bharosa Poora Na Ho

Level 3 सबसे tight security है। Isme upar bataye gaye approval तरीके (Live Confirm या Password) तो चाहिए ही होते हैं, साथ में ये तीन चीज़ें भी साथ चाहिए:

1. Apna password
2. Doosre user की approval
3. Local Device Verification Code

VERIFICATION CODE KAISE KAAM KARTA HAI

Ye code Internet, OTP, SMS, या Email — किसी पर भी depend नहीं करता। Ye tabhi generate होता है जब dono के devices एक doosre के **physical पास** हों (जैसे एक ही कमरे में, Bluetooth/Wi-Fi जैसी local range में)। अगर कोई door बैठाकर या internet के zariye access करने की कोशिश करे, तो code generate ही नहीं होगा। Code कुछ der में खुद expire भी हो जाता है।

Ye level उन लोगों के liye है जिनके पास बहुत sensitive data है, और जो एक doosre पर भी 100% bharosa नहीं करते — इसीलिए device की physical closeness एक extra layer बन जाती है।

6. Old Chat Principle

EXAMPLE

Socho aap Level 2 wale ek chat mein hain. Aapne apna password daal kar folder khol liya — lekin jo baatein pehle ho chuki thi, wo turant nahi dikhengi. Jaise hi doosra insaan (jiske saath chat hai) ijazat deta hai, tabhi wo purani baatein saamne aati hain. Aur jaise hi aap us screen se bahar nikalte ho — chahe app band karo ya kahin aur switch karo — abhi ki ye chat bhi turant "Old Chat" ban jaati hai. Matlab agli baar dekhne ke liye phir se doosre ki permission chahiye hogi.

Ye rule Level 2 aur Level 3 — dono mein lagu hota hai, aur ye ensure karta hai ki koi bhi purani sensitive baatchheet bina doosre ki marzi ke dobara na khul jaaye.

7. Hidden Privacy System

Framework mein Hidden Chat ka wajood kahin reveal nahi hota. Rules ye hain:

- Hidden Folder kahin visible nahi hota
- Uska koi icon nahi hota
- Uska koi button nahi hota
- App kabhi ye nahi batayega ki Hidden Folder pehle se exist karta hai

SABSE ZAROORI BAAT

Chahe user usi chat ko kitni bhi baar "Hide" karne ki koshish kare, app kabhi ye reveal nahi karega — "Already Hidden", "Already Exists", ya "Folder Created" jaisa koi bhi message kabhi nahi dikhega. Ek hi user ke liye unlimited Hidden Workspaces bhi ban sakte hain, aur ye sab bhi kabhi expose nahi hote.

8. Hidden Workspace Kaise Khole

- 1 Search Bar mein apna Hidden Password type karo
- 2 Search Button ko lagataar 3 baar tap karo
- 3 Password verify hoga
- 4 Workspace khul jayega

TRIPLE-TAP KYUN

Ye teen-baar-tap isliye rakha gaya hai taaki agar wahi password kisi normal chat mein bhi kabhi likha gaya ho, to sirf normal search karne se Hidden Workspace accidentally reveal na ho jaaye.

9. Decoy System

Advanced Mode mein ek hi Workspace, alag-alag password ke saath, alag-alag data dikha sakta hai.

EXAMPLE

Socho Priya ne apne workspace ke liye 3 password set kiye:

- **Original Password** — iske peeche uska asli, private data hai
- **Fake Password** — ek pehle se taiyar ki hui normal si chat, jisme harmless baatein hain
- **Empty Password** — bilkul khaali workspace

Agar kabhi koi Priya ko zabardasti phone unlock karne ko kahe, wo Fake Password de sakti hai. Us insaan ko lagega ki yehi original data hai — usse kabhi pata nahi chalega ki asli, private data kahin aur chhupa hua hai.

Password Type	Kya Dikhta Hai
Original	User ka asli, private data
Fake	Pehle se taiyar ki hui, believable chat — jise dekhne wala "original" samajhta hai
Empty	Bilkul khaali workspace; isme kiye badlaav session khatam hote hi apne aap hat sakte hain

10. Fake Workspace Principle

Original user ki pehli Fake Chat hi, us Temporary User (jisko wo password diya gaya) ke liye "Original Chat" ban jaati hai. Isi wajah se Temporary User ko kabhi ye pata nahi chalta ki wo asal mein ek Fake Layer dekh raha hai — uske liye wahi sach hai.

11. Password Management

Har user sirf apni *current* layer ka Original Password change kar sakta hai — kisi aur layer ka nahi.

EXAMPLE (SECTION 9 WALI KAHANI JAARI)

Priya ka asli password **1234** hai. Usne Fake Password **5678** bhi set kiya hua hai. Agar koi Temporary User 5678 se login karke apna password badal de, to sirf uski wahi Fake Layer update hoti hai — Priya ka asli 1234 password kabhi prabhavit (affect) nahi hota. Neeche wali layer, kabhi bhi upar wali layer ko modify nahi kar sakti.

12. Empty Workspace Rule

Empty Workspace har layer mein same tarah se kaam karta hai. Iska password kisi bhi layer ka current user badal sakta hai, kyunki Empty Workspace sabke liye ek common, neutral behaviour rakhta hai.

13. Recursive Privacy

SOCHNE KA AASAN TAREEKA

Isko Russian *Matryoshka* gudiya ki tarah socho — ek gudiya ke andar chhoti gudiya, uske andar aur chhoti. Bilkul waise hi, har Fake Workspace ke andar bhi apna Original, Empty, aur agli Fake layers ho sakti hain. Har layer, apne neeche wale insaan ke liye ek pura naya, believable "sach" bana deti hai — lekin koi bhi neeche wali layer, kabhi ye nahi jaan sakti ki uske upar aur bhi kuch chhupa hai.

14. Password Visibility Rule

User Kya Dekh Sakta Hai	User Kya Nahi Dekh Sakta
Apne se neeche wali layer mein hue password changes	Apne se upar wali layer ka Original Password — na dekh sakta hai, na modify kar sakta hai

15. Secure Communication Logs

Agar logging on hai, to messages kabhi seedhe, readable format mein store nahi hote. Har message ek internal Codec Language mein convert hokar save hota hai, aur ye Codec Pattern har session ke liye alag (dynamic) ho sakta hai. Isse agar koi log file bhi padh le, to asli conversation samajhna bahut mushkil ho jata hai.

16. Core Privacy Principle

Is framework ka maqsad sirf messages ko lock karna nahi hai. Asli maqsad ye hai ki:

- User khud decide kare data kab dikhe
- Kis ki permission se dikhe
- Kis level ki security mein dikhe
- Aur zarurat padne par, data ka wajood tak prove na kiya ja sake

17. Ek Nazar Mein Poora Framework

Feature	Normal Mode	Advanced Mode
Locked Folder	✓	✓
Hidden Folder (triple-tap)	✓	✓
Three Level Security	✗	✓
Decoy System (Original/Fake/Empty)	✗	✓
Recursive Privacy (layer ke andar layer)	✗	✓
Local Device Verification Code	✗	✓ (Level 3 mein)
Codec-based Secure Logs	✗	✓

18. Developer Ke Liye Notes

NOTE 1 — UNLIMITED HIDDEN WORKSPACES

Technically ek user ke liye unlimited Hidden Workspaces banana possible hai, lekin implementation karte waqt storage aur performance ki practical limits zaroor rakhein. Ye limit user ko dikhani zaroori nahi — ye sirf ek backend safeguard ki tarah kaam kar sakti hai.

NOTE 2 — RECURSIVE FAKE LAYERS

Recursive fake layers ka idea unique hai, lekin implementation aur testing bahut dhyan se karni hogi — kyunki har layer ka behaviour bilkul consistent rehna chahiye, warna kisi ek layer ke galat behave karne se poora deniability wala concept kamzor pad sakta hai.